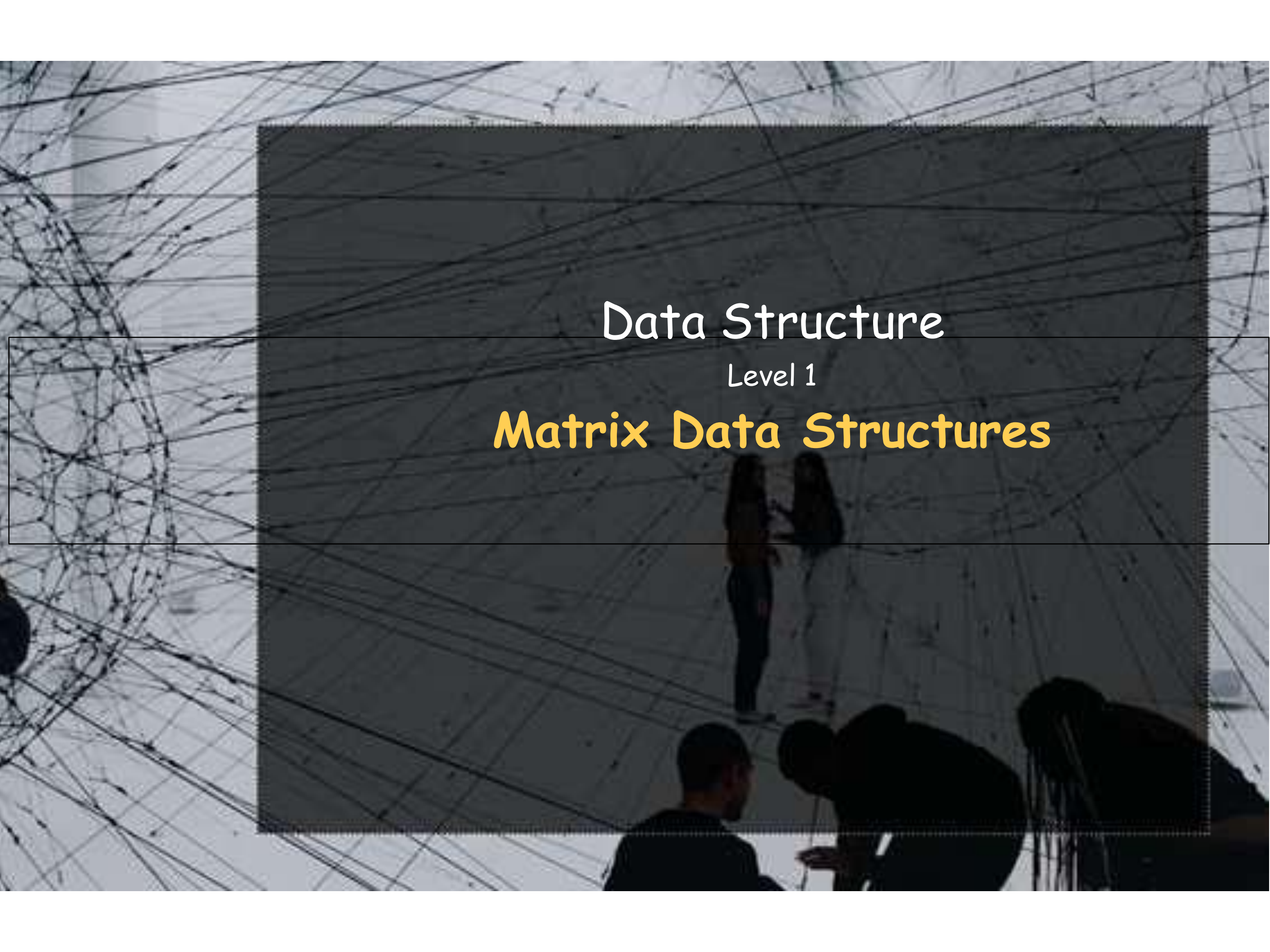


Data Structure

Level 1

Matrix Data Structures

The background of the slide features a complex, abstract network of thin, dark lines crisscrossing a light gray background, resembling a neural network or a data visualization. In the lower portion of the image, there are dark silhouettes of several people, some standing and some leaning over, suggesting a collaborative work environment or a meeting. A large, dark gray rectangular area is overlaid on the right side of the image, serving as a backdrop for the text.

Data Structure

Level 1

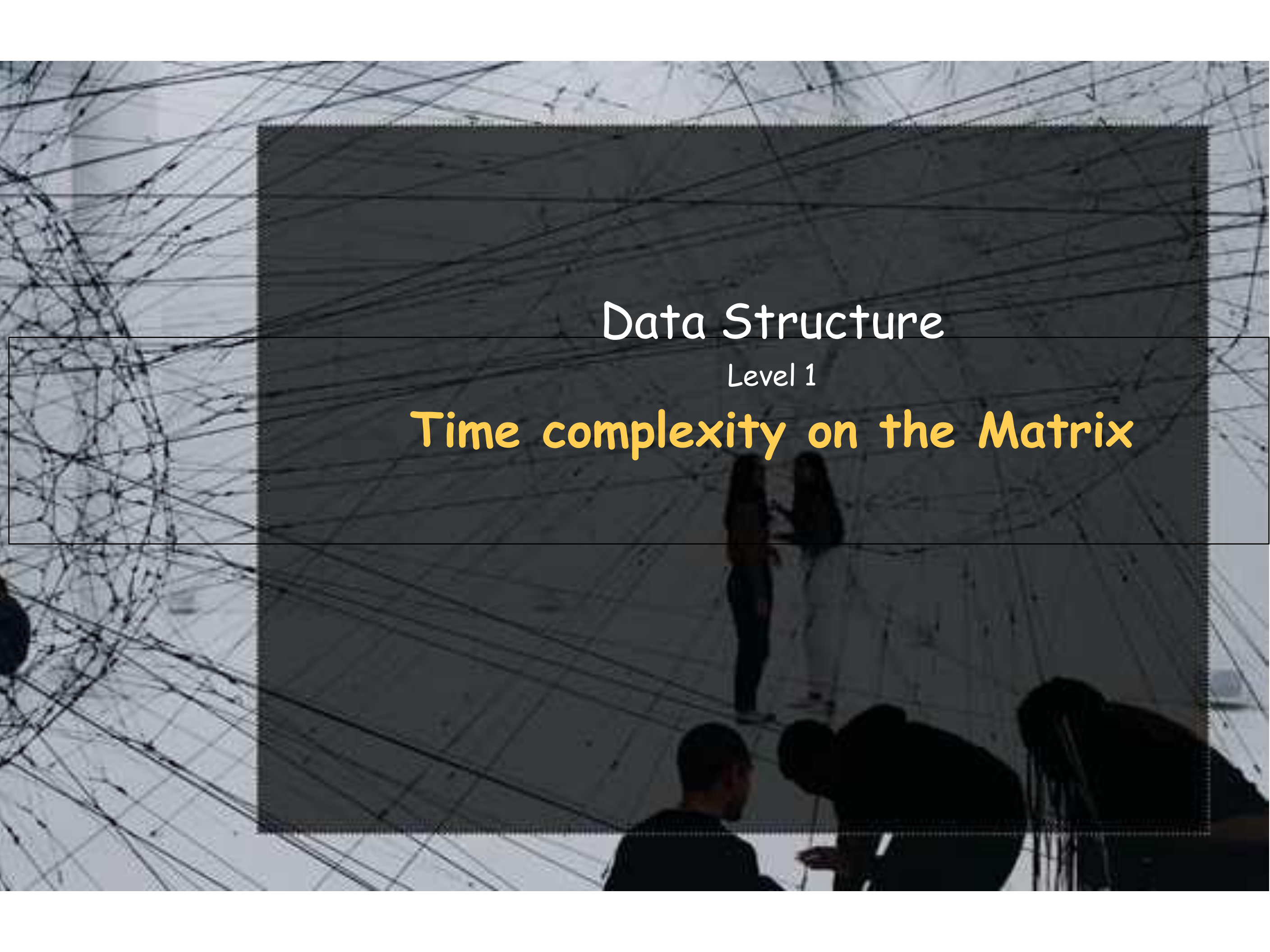
Matrix Data Structures

Matrix Data Structure:

- Matrix: A matrix represents a collection of numbers arranged in an order of rows and columns.

	Col 1	Col 2	Col 3	Col 4
Row 1	x[0][0]	x[0][1]	x[0][2]	x[0][3]
Row 2	x[1][0]	x[1][1]	x[1][2]	x[1][3]
Row 3	x[2][0]	x[2][1]	x[2][2]	x[2][3]

	Col 1	Col 2	Col 3	Col 4
Row 1	1	2	3	4
Row 2	5	6	7	8
Row 3	9	10	11	12

The background of the slide features a complex, abstract network of thin, dark lines crisscrossing a light gray background. In the lower portion of the image, there are dark silhouettes of several people, some standing and some appearing to be in motion, which adds a sense of activity and collaboration to the technical theme.

Data Structure

Level 1

Time complexity on the Matrix

Time complexity on the Matrix

■ Insertion: We try to insert a value to a particular array index position, as the array provides random access it can be done easily using the assignment operator.

💡 `arr[2][1] = 10;`

■ Time Complexity:

$O(1)$ to insert a single element

$O(N^2)$ to insert all the matrix elements [where N is the size of the array]

Time complexity on the Matrix

■ Access elements in Matrix: Accessing Matrix elements become extremely important, in order to perform operations on arrays.

💡 `return arr[2][1];`

■ Time Complexity:

$O(1)$ to insert a single element

$O(N^2)$ to insert all the matrix elements [where N is the size of the array]

Time complexity on the Matrix

- Searching in Array: We try to find a particular value in the matrix, in order to do that we need to access all the matrix elements and look for the particular value.

```
// searching for value 55 in the matrix;
```

```
Loop from i = 0 to 10:
```

```
    Loop from j = 0 to 10:
```

```
        check if arr[i][j] = 55:
```

```
            return true;
```

- Time Complexity:
 $O(N^2)$ [where N is the size of the array]

Thank you 😊